

Swarnav Mukherjee

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EDUCATION

Netaji Subhas University of Technology (NSUT)

New Delhi, India

Bachelor of Technology in Electronics and Communication Engineering (ECE)

2023 – 2027

- Relevant Coursework: Data Structures & Algorithms, Microprocessor and Computer Architecture, Embedded Systems, Electronic Devices and Circuits, Digital Logic Design

EXPERIENCE

Vayumandal Innovations

Remote

Research and Development Intern

September 2025 – November 2025

- Supported the design, prototyping, and refinement of embedded electronics for drone systems
- Conducted schematic analysis, circuit optimization, and initial testing and validation processes, bridging the gap between hardware layout and functional performance.
- Researched UAV Swarm Technology, focusing on control logic architectures, decentralized communication frameworks, and multi-drone coordination for real-world applications

PROJECTS

Custom Unix Shell | C++, Linux API, Process Management

In Progress

- Developed a functional command-line interface (CLI) shell in C++ capable of executing system commands.
- Implemented core process management using `fork()`, `execvp()`, and `waitpid()` system calls.
- Engineered input parsing logic to handle command arguments and managed foreground/background process execution.
- Demonstrated low-level memory management and direct interaction with the Linux Kernel API.

Turbotyping – Real-Time Typing Speed Platform | React, Express.js, Socket.io, MongoDB, TailwindCSS.

- Engineered a real-time typing platform with single-player and multiplayer modes using **WebSockets** (Socket.io); maintained **sub-100ms** sync for live leaderboards and opponent cursors.
- Implemented keystroke-level telemetry to compute WPM and accuracy, persisted per-user stats in **MongoDB**, and provided endpoints for leaderboard aggregation and historical trends.
- Built modular test generators (timer/word modes) with dynamic prompts and robust anti-cheat checks; enabled authenticated sessions with JWT.

High-Performance TCP Server | C++, POSIX Sockets, Linux

- Developed a concurrent TCP server from scratch using C++ and POSIX sockets to reliably handle multiple simultaneous client connections.
- Leveraged Linux system calls and `epoll` for efficient I/O multiplexing, optimizing transport layer network traffic and minimizing response latency.
- Implemented robust error handling, custom stream buffering, and strict memory management to ensure stable, long-running application performance without memory leaks.

TECHNICAL SKILLS

Languages: C++, System Verilog, C, JavaScript (ES6+), Python, SQL

Systems & Hardware: Linux/Unix, Operating Systems, Computer Architecture, Digital Logic, ESP32, Arduino

Web Technologies: React, Redux Toolkit, Node.js, Express.js, REST APIs, WebSockets, HTML/CSS

Tools: Git, Docker, VS Code, EDA Playground, Postman, MongoDB

ACHIEVEMENTS

JEE Mains: Secured 97.4 percentile in JEE Mains, demonstrating strong aptitude in Math and Physics.

Competitive Programming: Solved over 500 Questions on Leetcode and Codeforces